INTEGER WEIGHTS NEVER GIVE A SOUND COLLESS-LIKE BALANCE INDEX

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Abstract. Mir, Rotger and Rosselló introduced Colless-like balance indices $\mathfrak{C}_{D,f}$ for rooted multifurcating trees, depending on a dissimilarity D and a weight function f on outdegrees, and conjectured that no f with values in the natural numbers makes $\mathfrak{C}_{D,f}$ sound, that is, zero exactly on fully symmetric trees. We prove the conjecture in a stronger form: soundness fails whenever $f(2), f(3), f(4)$ are rational multiples of one real number, whatever the other values of f. The proof counts fully symmetric trees with outdegrees in $\{2,3,4\}$: there are about 27^m of them with 24^m leaves, while, after removing their common leaf contribution and scaling, the internal f-sizes occupy only $O(24^m)$ integer values.

1. The conjecture

All trees are rooted, without vertices of outdegree one. For f mapping outdegrees to non-negative reals, the *f-size* of a tree is $\delta_f(T) = \sum_v f(\deg^+ v)$ over all vertices. For a dissimilarity D (a symmetric function of finite lists of reals that vanishes exactly on constant lists), the Colless-like index is

$$\mathfrak{C}_{D,f}(T) = \sum_{v \text{ internal}} D\big(\delta_f(T_{v_1}), \dots, \delta_f(T_{v_k})\big),$$

where $T_{v_1}, \dots, T_{v_k}$ are the subtrees rooted at the children of v [2]. A tree is *fully symmetric* if at every internal vertex all child subtrees are isomorphic, and $\mathfrak{C}_{D,f}$ is *sound* if it vanishes exactly on fully symmetric trees. Mir, Rotger and Rosselló proved that $\mathfrak{C}_{D,f}$ is sound if and only if δ_f separates non-isomorphic fully symmetric trees [2, Lemma 10], showed soundness for $f(n) = e^n$ and $f(n) = \ln(n+e)$, and ended with: "Our conjecture is that there is no function f taking values in the set of natural numbers that satisfies this property" [2]. The survey of Fischer, Herbst, Kersting, Kühn and Wicke lists the conjecture as open [1, Sec. 9.7].

Theorem 1.1. *Let f be a weight function and suppose $f(2), f(3), f(4) \in \gamma\mathbb{Z}$ for some real $\gamma > 0$, for instance all rational. Then there are two non-isomorphic fully symmetric trees with the same f-size, so $\mathfrak{C}_{D,f}$ is not sound for any dissimilarity D. In particular the conjecture of Mir, Rotger and Rosselló holds.*

2. Proof

A fully symmetric tree is determined by its sequence of outdegrees level by level, $w = (k_1, \dots, k_h)$ with $k_i \geq 2$; write FS_w, and note that different words give non-isomorphic trees. With prefix products $p_0 = 1$ and $p_i = k_1 \cdots k_i$, level $i-1$ has p_{i-1} vertices of outdegree k_i and there are p_h leaves, so

$$\delta_f(\mathrm{FS}_w) = \sum_{i=1}^{h} p_{i-1} f(k_i) + p_h f(0).$$

Write $f(j) = \gamma a_j$ with integers $a_j \geq 0$ for $j \in \{2,3,4\}$ and let $M = \max_j a_j$. For $m \geq 1$ let W_m be the set of words with exactly m letters equal to each of $2, 3, 4$. Every $w \in W_m$ has $p_h = 24^m$, so the leaf term $24^m f(0)$ is the same for all of them, and two words have equal f-size exactly when their internal sums $\sum_{i=1}^{h} p_{i-1} f(k_i)$ agree. Since $1 + p_1 + \cdots + p_{h-1} < p_h$ for words with

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letters at least 2 [2, Lemma 15], the internal sum divided by γ is one of the $M \cdot 24^m + 1$ integers in $[0, M \cdot 24^m]$. On the other hand $|W_m| = (3m)!/(m!)^3$ is the largest of the $\binom{3m+2}{2}$ multinomial coefficients summing to 27^m, so

$$|W_m| \;\geq\; \frac{27^m}{\binom{3m+2}{2}} \;>\; M \cdot 24^m + 1$$

for all large m, because $(27/24)^m$ grows exponentially. By pigeonhole two distinct words in W_m have equal f-size. If FS_w and $\mathrm{FS}_{w'}$ are such trees, the tree whose root has exactly these two children has $\mathfrak{C}_{D,f} = 0$ (every internal vertex inside either child is balanced, and the root sees two equal f-sizes) but is not fully symmetric. □

3. Remarks

Small examples. The pigeonhole bound guarantees collisions only for enormous trees (for weights up to 50, at $m \approx 80$), but natural integer weights collide immediately. With $f(n) = n+1$, $\mathrm{FS}_{(2,2)}$ and $\mathrm{FS}_{(6)}$ both have f-size 13; with $f(n) = 2^n$, $\mathrm{FS}_{(2,3)}$ and $\mathrm{FS}_{(3,2)}$ have 26; with $f(n) = n! + 1$, $\mathrm{FS}_{(4,3)}$ and $\mathrm{FS}_{(2,2,2,2)}$ have 77; with $f(n) = n^3 + 1$, $\mathrm{FS}_{(3)}$ and $\mathrm{FS}_{(2,2)}$ have 31. A standard-library script checks these by building the trees and evaluating $\mathfrak{C}_{D,f}$ for the variance, the standard deviation and the mean deviation from the median.

Where the method stops. This fixed-alphabet argument does not settle weights in a rank-two group such as $\mathbb{Z} + \mathbb{Z}\sqrt{2}$. If $f(2), f(3), f(4)$ are integer combinations of two fixed generators, the internal sums have integer-coordinate pairs in a box with $O(24^{2m})$ points. This exceeds the approximately 27^m words used here, so this comparison no longer forces a collision. It is a limitation of this argument, not a proof that no other counting argument works. Rank-two weights can collide: Mir, Rotger and Rosselló give $\delta_f(\mathrm{FS}_{8,10}) = \delta_f(\mathrm{FS}_{12,8}) = 352$ for $f(n) = (\sqrt{2})^n$ [2]. Whether some weight function with values in a rank-two group is sound is not settled here.

Domain. The original paper writes $f\colon \mathbb{N} \to \mathbb{R}_{>0}$ but uses $f(0)$ and zero values in examples; the survey [1] takes $f\colon \mathbb{N}_{\geq 0} \to \mathbb{R}_{\geq 0}$. The theorem covers every such convention, since it only uses $f(2), f(3), f(4)$ and neither positivity nor the value $f(0)$.

Contributions and AI use

Harvey Yorke directed the project. Coding agents developed the arguments, the verification checkers and the manuscript drafts. Valyu DeepResearch carried out the literature, priority and internal adversarial reviews. These reviews are internal; the manuscript has not undergone external peer review. Harvey Yorke has not yet completed a line-by-line review of every proof; correctness rests on the written proofs and on the checks and any formal verification described in this paper.

References

[1] Mareike Fischer, Lina Herbst, Sophie Kersting, Annemarie Luise Kühn, and Kristina Wicke. *Tree Balance Indices: A Comprehensive Survey*. Springer, Cham, 2023. arXiv:2109.12281v2.

[2] Arnau Mir, Lucía Rotger, and Francesc Rosselló. Sound Colless-like balance indices for multifurcating trees. *PLoS ONE*, 13(9):e0203401, 2018. arXiv:1805.01329.